

The POST error message “**CMOS checksum error – Defaults loaded**” means that the BIOS detected that the stored BIOS configuration settings in the **CMOS memory** are incorrect or corrupted. The checksum value (a small data verification value used to confirm the settings are intact) does not match, so the system assumes the settings may be unreliable and automatically loads the default BIOS settings.

Common causes:

- A **weak or dead CMOS battery** (most common cause)
- BIOS settings being reset after power loss
- Incorrect BIOS configuration changes
- Corrupted BIOS settings

How to resolve it in a real system:

1. Replace the CMOS battery

- Shut down the PC and disconnect power.
- Open the case and replace the motherboard’s coin-cell battery (commonly a **CR2032**).
- Restart the computer and check if the error disappears.

2. Reconfigure BIOS settings

- Enter the BIOS/UEFI setup (usually by pressing **Delete, F2, or another key shown during startup**).
- Confirm settings such as **date/time, boot order, and hardware settings**.
- Save the changes and restart the system.